MILITARY SPECIFICATION

PRESSURE CARTRIDGE PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection of the parts, loading, assembling and packing for one type of electrically initiated pressure cartridge.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	Box, Wood, Nailed and Lock Corner

MILITARY

MIL-S-45743	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment
MIL-B-117	Bags, Sleeves and Tubing - Interior Packaging
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging of
MIL-A-48078	Ammunition Standard Quality Assurance Provisions, General Specification for

FSC : 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) a ppearing at the end of this document or byhtter.

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STANDARDS

MI LX TARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Snipnent and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-650	Explosive, Sampling, Inspection Testing

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT C O W D

9292613	Pressure Cartridge

(Cop~es of specifications, standards, drawings, and publications required by suppl~ers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, '49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on Drawing (Dwg.) 9292613, all associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Moisture content of intermediate charge (calcium chromate and amorphous boron powder) . The moisture content of the calcium chromate - boron powder intermediate charge at the loading station at the time of loading shall not exceed 0.50 percent.

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3.4 Functioning output pressure/time test. The pressure cartridge shall function on the first attempt and shall produce a pressure of 450 psig minimum (rein) for a time duration of 75 milliseconds (ins), when fired in a T-bomb with an internal volume of 0.049 cubic inches (in^3). The time to peak pressure shall be 50 ms maximum (max) after application of the ignition pulse and the peak pressure shall be 2400 psig max (see fig 1). The cartridge shall function when temperature conditioned as specified in 4.5.8.

3.5 Insulation resistance.

3.5.1 Insulation resistance of pressure cartridge assembly. The insulation resistance between the shorted (twisted) electrodes and the case shall be 50 megohms rein, at 325 volts direct current (vdc) when tested as specified in 4.5.4.

3.5.2 Insulation resistance, thermal shock and vibration and function. The pressure cartridge shall comply with the insulation resistance requirements of 3.5.1, and shall function on the first attempt when tested as specified in 4.5.5.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated, and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Case	9292661	25
Disk, Closure	9292663	25
Disk, Screen	9292664	25
Sleeve, Header	9292665	25
Bridged Header and Ignition Assembly	9292666	25
Header, Bridged	9292667	50
Header	9292668	25
Sleeve	9292670	25
Pressure Cartridge	9292613	225

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Pressure Cartridge and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case				
	(Dwg. 9292661)				
	Examination for defects	25		3.2	4.4.2.1
	Disk, Closure				
	(Dwg. 9292663)				
	Examination for defects	25		3.2	4.4.2.2
	Disk, Screen				
	(Dwg. 9292664)				
	Examination for defects	25		3.2	4.4.2.3
	Sleeve, Header				
	(Dwg. 9292665)				
	Examination for defects	25		3.2	4.4.2.4
	Header				
	(Dwg. 9292668)				
	Examination for defects	25		3.2	4.4.2.5
	Pull Test	25		3.2	4.5.2
	Sleeve				
	(Dwg. 9292670)				
	Examination for defects	25		3.2	4.4.2.8

NOTES:

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TABLE I FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Pressure Cartridge and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Header, Bridged (Dwg. 9292667)				
	Examination for defects	50		3.2	4.4.2.6
	Bridgewire Resistance Test	50		3.2	4.5.3
	Bridged Header and Ignition Assy. (Dwg. 9292666)				
	Examination for defects	25		3.2	4.4.2.7
	Pressure Cartridge (Dwg. 9292613)	225			
	Examination for defects	(a)		3.2	4.4.2.10
	Bridgewire Resistance	(a)		3.2	4.5.3
	Insulation Resistance	(a)		3.5.1	4.5.4
	Insulation Resistance, thermal shock, vibration, function	25	0-1	3.5.2	4.5.5
	No-fire	200	0-1	3.2	4.5.6
	Output pressure/time test				
	Cold (-35°F -5°F)	100(b)	0-1	3.4	4.5.8.1
	Hot (145°F +5°F)	100(b)	0-1	3.4	4.5.8.2

NOTES:

(a) The entire quantity of pressure cartridges shall be subjected to the examination/test specified prior to testing below

(b) Samples for this test shall be taken from no-fire test samples.

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4.4 Quality conformance Inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of pressure cartridges shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.
b. Lead styphnate from not more than one lot.
c. Intermediate charge (calcium chromate and amorphous boron powder) from not more than one lot.
d. Smokeless powder from not more than one lot.

4.4.2 Examination. See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63315(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Case				9292661
					NEXT HIGHER ASSEMBLY 9292613
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness at bottom		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Diameter of hole in bottom		0.40%	3.2	Gage
105	Temper improper		-	3.2	Certification
Minor					
201	Overall length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Disk, Closure				9292663
					NEXT HIGHER ASSEMBLY
					9292613
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Disk, Screen				9292664
					NEXT HIGHER ASSEMBLY 9292613
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63315(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Sleeve, Header				9292665
					NEXT HIGHER ASSEMBLY
					9292613
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Length		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-83315(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Header				9292668
					NEXT HIGHER ASSEMBLY 9292667
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Center to center spacing of electrodes		0.40%	3.2	Gage
102	Surface finish (before plating) of top of header		0.40%	3.2	Gage/Witness blocks
103	Pull Test of electrodes		50-0-1	3.2	4.5.2
104	Solderability of electrode, 2 places	3	0-1	3.2	MIL-S-45743
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

a. Sample as per MIL-STD-105, Level S-3.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Header, Bridged				9292667
					NEXT HIGHER ASSEMBLY 9292666
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridgewire material improper		0.40%	3.2	Certification
102	Bridgewire weld missing or improper (2 places)		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Bridged Header and Ignition Assembly	1 OF 1	9292666
			NEXT HIGHER ASSEMBLY: 9292613

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead styphnate on outside diameter of header		100%	3.2	Visual
102	Potting missing or improper		0.40%	3.2	Visual
Minor					
201	Length of lead insulation		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63315(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Sleeve				9292670
					NEXT HIGHER ASSEMBLY
					9292668
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Gage
202	Wall thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63315(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Pressure Cartridge (prior to shorting)		9292613
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Bridgewire resistance		100%	3.2	4.5.3
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.10	Pressure Cartridge	3	9209C13
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		[illegible]	3.2	[illegible]
102	Outside diameter (for specified length		[illegible]	[illegible]	[illegible]
103	Bare lead wires not shorted (twisted) together		100%	3.2	Visual
104	Assembly cracked, punctured or split		100%	3.2	Visual
105	Cup crimp improper		0.40%	3.2	[illegible]
106	Sealant over closure disk missing or improper		[illegible]	[illegible]	[illegible]
107	Epoxy adhesive at cup crimp missing or improper		[illegible]	3.2	Visual
108	Insulation resistance		[illegible]	[illegible]	[illegible]
109	Insulation resistance, thermal shock vibration and function	[illegible]	0-1	3.5.2	4.5.2
110	No-fire test	[illegible]	0-1	3.5	4.5.6
111	All-fire test	[illegible]	0-1	3.2, 3.4	4.5.7
112	Output pressure/time test				
	Cold temperature (-35°F - 5°F)	[illegible](a)	(b)	3.4	4.5.8.1
	Hot temperature (145°F + 5°F)	80(a)	(b)	3.4	4.5.8.2
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES a. Samples for this test shall be taken from pressure cartridge subjected to the No-fire test.
b. Failure to function on first attempt, 80-0-1 Pressure/time failure, 80-1-2.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Packaging, Packing and Marking				See Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Partitioning or cushioning improper		0.40%	3.2	5.1/Visual
102	Barrier bag torn, punctured or sealed improperly		0.40%	3.2	5.1/Visual
103	Outer box damaged or not properly secured		0.40%	3.2	5.2/Visual
Minor					
201	Marking missing, illegible or improper		0.65%	3.2	5.3/Visual
202	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Con formance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment requi red to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval Inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Moisture content of intermediate charge (calcuim chromate and amorphous boron powder). The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture content of the intermediate charge s'nail be determined at the loading station at the time of loading. This determination shall be made at least once each four (4) hours of production. Sampling of the intermediate charge shall be in accordance with MIL-STD-650, Method 101.5. If the moisture content fails to comply with the applicable requirements, the intermediate charge material shall be rejected. If loading has begun, pressure cartridges loaded with the non-conformance explosive charge shall 'be held in reject status until corrective action has been verified.

4.5.2 Pull test of electrodes of header. The header shall be assembled in an approved fixture and the procedure of MIL-STD-202, Method 211, Test Condition A shall be used with the exception that the force specified on the drawing shall be applied to each electrode for 45 seconds. The force shall be a gradually applied static axial load. Any header in which either electrode moves or loosens shall be classed defective. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.3 Bridgewire resistance of pressure cartridge. Prior to shorting (twisting) the electrodes, the assembly shall be connected to approved equipment and the bridgewire (electrode to electrode) resistance measured to determine compliance with the requirement specified on the applicable drawing. Any pressure cartridge failing to comply with the applicable drawing requirement shall be classed defective and removed from the lot. The measurement shall be made with a meter capable of measurement to the nearest 0.01 ohm. The current applied shall be limited to one (1) milliampere, max. (Non-destructive test).

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4.5.4 Insulation resistance of pressure cartridge. The insulation resistance between the shorted (twisted) electrodes and the case shall be determined using the procedure specified in MIL-STD-202, Method 302. The specified voltage shall be applied for 2 seconds minimum. Examination shall be made to determine compliance with 3.5.1 as well as any evidence of insulation failure such as loosening, cracking, or charring of the insulation tubing or reduced voltage. Any cartridge which fails to comply with the applicable requirement shall be classed defective. (Non-destructive test)

4.5.5 Insulation resistance, thermal shock, vi bration and function. The pressure cartr idges shall be tested as follows, in the sequence specified, using approved equipment:

a. Measure resistance between the shorted electrodes and case in accordance with 4.5.4.
b. Condition for two (2) hours minimum at minus 65 ± 5°F.
c. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 165 ± 5°F.
d. Condition for two (2) hours minimum at 165 + 5°F and 20% maximum relative humidity.
e. Remove from hot chamber.
f. Vibrate for 15 minutes, minimum in a sweep covering a frequency range of 25-500-25 cycles per second, parallel to the transverse orthogonal axis, with output and down and a 2 + 0.2 G's peak amplitude.
g. Measure resistance between shorted electrodes and case in accordance with 4.5.4.
h. The pressure cartridges shall be tested, as specified in 4.5.6 and 4.5.7. Any pressure cartridge which fails to comply with the applicable requirement shall be classed defective. (Destructive test)

4.5.6 No-fire voltage of pressure cartridge. The samples for this test shall be selected after the performance of bridgewire resistance of 4.5.3. The pressure cartridge shall be subjected to the specified no-fire voltage by discharging a capacitor across the bridge circuit in accordance with the requirements of Dwg. 9292613. The bridgewire resistance shall be checked as specified in paragraph 4.5.3 to assure that the bridgewire has remained intact. Any sample which fails to comply with the applicable drawing requirement shall be classed defective and shall cause the lot to be rejected. A record of each firing shall be kept. See 6.7. (Non-destructive test).

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4.5.7 All-fire voltage of pressure cartridge (70°F + 10°F). The samples for this test shall be selected after satisfactory performance of the no-fire voltage test of 4.5.6. The pressure cartridge shall be subjected to the specified all-fire voltage by discharging a capacitor across the bridge circuit in accordance with the requirements of Dwg. 9292613. Observation shall be made for compliance with the applicable drawing requirement. Any sample which fails to fire on the first attempt shall be classed defective. A record of each firing shall be kept. See 6.7. (Destructive test).

4.5.8 Output pressure/time test of pressure cartridge.

4.5.8.1 Cold temperature (minus 35°F - 5°F). The pressure cartridge shall be assembled in a fixture with the configuration of Fig. 2 which simulates the volume of the sensor system as specified in paragraph 3.4 and which is instrumented to provide a record of pressure versus time. The test samples and fixture shall be conditioned at the required test temperature for two (2) hours minimum. The pressure cartridge shall be initiated at the test temperature using the all-fire voltage specified on Dwg. 9292613, and the pressure/time results evaluated against the requirement specified in 3.4. If any pressure cartridge fails to function and/or more than one (1) pressure/time failure occurs, the lot shall be rejected. Care shall be taken during conditioning and handling of the assembles to avoid contamination. The fixture shall be cleaned after each shot, or as requried. A record of each firing shall be kept. See 6.7. (Destructive test)

4.5.8.2 Hot temperature (+145°F + 5°F). The pressure cartridge shall be tested using the procedure specified in 4.5.8.1 except that the conditioning temperature shall be as required. Observation shall be made for compliance with the requirements specified in 3.4. If any pressure cartridge fails to function, and,/or more than one (1) pressure/tune failure occurs, the lot shall be rejected. (Destructive test) . A record of each firing shall be kept. See 6.7.

5. PACKAGING

5.1 Packaging. The pressure cartaridge shall be packaged in accordance with level B of Spec MIL-P-14232 with the following mandatory requirements. The cartridges shall be packaged in a fiberboard or other rigid box with separators, loose partitions, or trays as required. Cushioning as needed shall be used. The box shall be sealed in a barrier bag conforming to type I, class F, style 1, Spec MIL-B-117.

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5.2 Packing. The pressure cartridge shall be packed in accordance with level B of Spec MIL-P-14232 and Code of Federal Regulations, Title 49 with the following mandatory requirements. The outer packing box shall be wood conforming to class 1, style 4, Spec PPP-B-621 or fiberboard conforming to Spec 12H, 23F, or 23H CFR Title 49. Sufficient cushioning and separators must be used between the barrier bag and outer box to prevent damage to the items or tearing of the barrier bag.

5.3 Marking. The marking of the outer box shall be in accordance with MIL-STD-129 and DOT marking per Code of Federal Regulation, Title 49.

5.4 Shipping. When end items from more than one lot are shipped at one time, such as in a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The components *covered* by this specification are intended for use in the:

a. XM74 AP mine
b. BLU-92/B AP mine

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Distribution of ammunition data cards. Dlstribution of data cards shall include the following: Commander , ARRADCOM, ATTN: DRDAR-QAA-Q, Dover, NJ 07801

6.5 Check test for deterioration of pressure cartridge assemblies. The following test is designed to prevent presure cartridges that nave deteriorated from being assembled into major ammunition items. The test shall be performed on pressure cartridges from each lot which have been stored for more than two (2) years, under adverse conditions for any period of time, or at the request of the Contracting Officer. If the specification for the ammunition item into which the pressure cartridges are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

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6.5.1 Functioning. One hundred sixty (160) pressure cartridges shall be selected from each lot and subjected to the hot and cold functioning test specified in 4.5.8; eighty (80) at each temperature. If two (2) or more of the eighty pressure cartridges fail to comply with the applicable cold or hot functioning requirements, including duds, the lot shall be rejected. If any lot is rejected, all data generated shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAA-Q, Dover, NJ 07801 (Destructive test).

6,6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

6.7 Data submission. Results of all tests performed by the contractor including copies of pressure/time traces shall be submitted in accordance with Data Item DI-R-1721 on the DD Form 1423 for the contract. Distribution shall include the following:

Commander
ARRADCOM
ATTN: DRDAR-QAA-Q
Dover, NJ 07801

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No: 1345-A-224

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T1 = Time Pressure First Reaches 450 psi

T2 = T1 + 75 MSEC

Point A Must Be No Earlier Than T2

Tp = 50 MSEC Max.

NOTE: Not To Scale

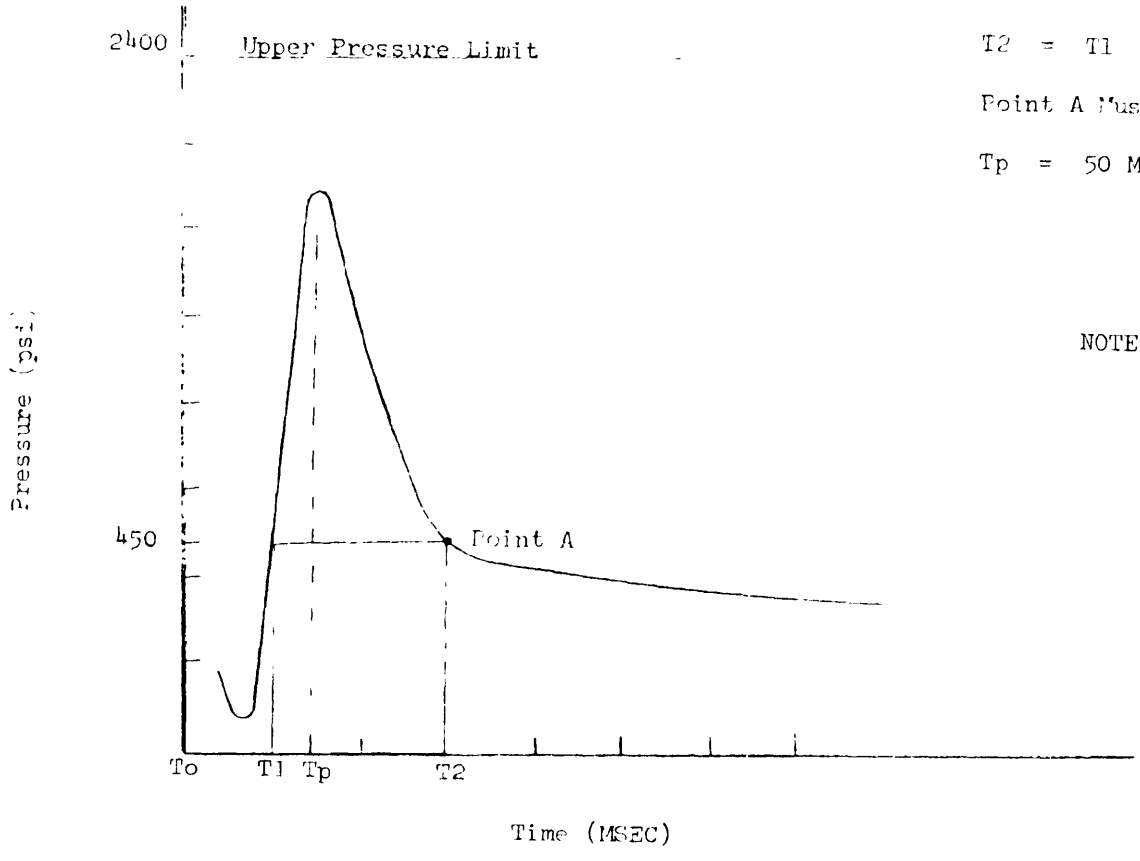
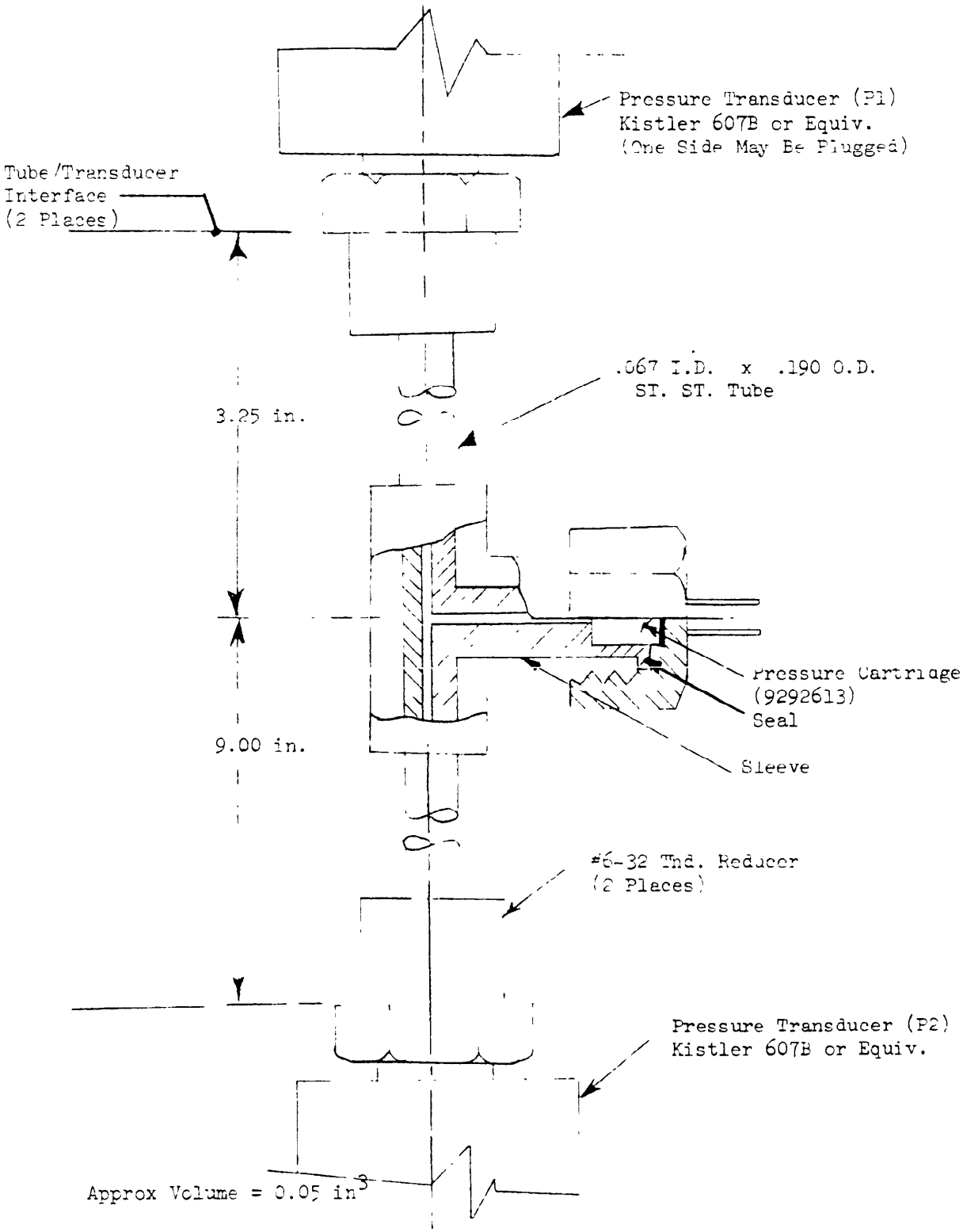


FIG. 1 PRESSURE CARTRIDGE REQUIREMENTS

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FIGURE 2. PRESSURE TEST SETUP (SUGGESTED)

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PENALTY FOR PRIVATE USE $300

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAA
Dover, New Jersey 07801

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DOCUMENT IDENTIFIER (Number) AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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